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PROJECT AND TEAM INFORMATION

Project Title

GigConnect- Bridging skills and opportunities

Student/Team Information

Team Name: ***Team: T-203***	***Raptors***
Team member 1 (Team Lead) *(Last Name, name: student ID: email, picture):*	***Suyal, Harshit – 230213596*** *harshitsuyal2000@gmail.com*
Team member 2 *(Last Name, name: student ID: email, picture):*	***Joshi, Manas – 230211450*** *jmanas275@gmail.com*

Team member 3 *(Last Name, name: student ID: email, picture):*	***Singh, Saumya Pratap – 23022353*** saumya246761@gmail.com
Team member 4 *(Last Name, name: student ID: email, picture):*	***Bisht, Aishwary – 230211549*** ***bishtvaibhav2@gmail.com***

PROJECT PROGRESS DESCRIPTION

Project Abstract

This project is to create a web-based platform that connects clients with gig workers in a simple, secure, and efficient way. Gig jobs, where people work on short-term tasks, are becoming more popular, but the process of finding and managing such work is often complex, expensive, and lacks transparency.

The platform solves these issues by including features like user authentication, easy job posting, application tracking, and dashboards created to satisfy the needs of both clients and workers. With a focus on being user-friendly and accessible, it eliminates high fees and simplifies communication between both parties.

This system is designed to make gig work more organized and reliable, providing a safe space for workers and clients to collaborate while offering tools to manage tasks easily. It ensures flexibility and convenience, making it a valuable solution for small businesses and people looking for temporary work opportunities.

Updated Project Approach and Architecture

1. **Core System Development:**
   - Frontend: Built with React.js to provide a responsive and dynamic user interface.
   - Backend: Developed using Node.js with Express.js for server-side logic.
   - APIs: RESTful APIs for communication between the frontend and backend.
   - Database: MongoDB is used to manage data for users, jobs, and applications.
   - Architecture: An MVC-based structure ensures clean and organized code.

2. **User Authentication & Authorization:**
   The platform prioritizes secure and role-based access to its features:
   - Users can register, log in, and manage accounts with password encryption.
   - JWT (JSON Web Token) is used for session management.
   - Role-based access is implemented for Admin, Clients, and Workers.

3. **Job Management Module:**
   This module simplifies client and worker interactions:
   - Clients can post, update, and delete job listings.
   - Workers can search, filter, and apply for jobs.

- Application tracking system allows status updates (e.g., Pending, Accepted, Rejected).
- Dashboards provide an overview of job-posting and application activities.

4. **Rating & Feedback System:**
To ensure accountability and transparency:
- Clients rate workers after job completion, and workers give feedback on clients.
- User profiles display average ratings to show reliability.

5. **Security & Data Handling:**
- Middleware-secured API endpoints and input validation prevent vulnerabilities.
- A structured data schema ensures efficient querying and storage.
- HTTPS is used for secure communication during deployment.

Tasks Completed

Task Completed	Team Member
1. Designed UI layouts and implemented responsive frontend using React.js	Manas Joshi, Saumya Pratap Singh
2. Developed backend APIs using Node.js and Express.js	Harshit Suyal, Manas Joshi
3. Implemented MongoDB database schema and integration	Saumya Pratap Singh, Aishwary Bisht
4. Built user authentication system (JWT, login/signup, roles)	Aishwary Bisht, Harshit Suyal
5. Implemented job posting and application features	Entire Team
6. Complete Rating & Feedback system implementation	Manas Joshi, Harshit Suyal
7. Implement secure Payment Integration module	Harshit Suyal, Aishwary Bisht
8. Enhance Job Management Module	Aishwary Bisht, Saumya Pratap Singh

9. Perform end-to-end testing and bug fixing	Entire Team
10. Optimize UI/UX and improve responsiveness further	Saumya Pratap Singh, Manas Joshi

Challenges/Roadblocks

1. **Frontend Responsiveness:**
   - Challenge: Making the user interface work smoothly across different devices and screen sizes.
   - Solution: Creating a simple, flexible design and testing it manually on various devices to ensure consistency.

2. **API Integration Issues:**
   - Challenge: The frontend and backend didn't always sync well during the initial stages.
   - Solution: Improved communication within the team to ensure both sides understood the requirements clearly and fixed issues together.

3. **Secure User Authentication:**
   - Challenge: Implementing a safe and reliable authentication system.
   - Solution: Added password encryption and created clear role-based access controls for different user types.

4. **Application Tracking Flow:**
   - Challenge: Defining and managing the statuses (Pending, Accepted, Rejected) for job applications properly.
   - Solution: Discussed and finalized a logical flow as a team and implemented it step by step.

5. **Team Collaboration:**
   - Challenge: Coordinating tasks effectively in a team with different roles.
   - Solution: Regular team discussions and updates helped to keep everyone on the same page.

6. **Debugging Problems:**
   - Challenge: Identifying and fixing small mistakes that caused bigger issues in the system.
   - Solution: Worked together as a team, testing and checking every component to resolve errors patiently.

Project Outcomes

1. **Secure User Management**
   - Authentication and authorization system.
   - Role-based dashboards for different users.

2. **Efficient Job Matching**
   - Structured job posting and application workflow.
   - Search and filtering mechanisms for improved usability.

3. **Transparent Interaction**
   - Rating and feedback system for trust building.
   - Clear job status tracking.

4. **Scalable Full Stack Architecture**
   - RESTful backend services.
   - Responsive frontend interface.

5. **Performance Optimization**
   - Optimized database queries for faster data retrieval.
   - Efficient API handling for reduced server load.

Progress Overview

The project has been successfully completed, with all planned functionalities fully implemented and operational. The frontend development, backend implementation, database design, and user authentication systems were completed as scheduled and are functioning effectively, providing a strong foundation for the platform.

The Job Management Module has been fully developed, including features such as job posting and application tracking. Additionally, the Rating and Feedback system and Payment Integration have been successfully implemented, completing all remaining components of the project.

Comprehensive testing and debugging have been carried out to ensure that all modules work seamlessly together. Any delays encountered during development were effectively managed, allowing the team to deliver the project without impacting the final deadline.
Overall, the project has been completed, meeting all objectives and requirements, with all features working as intended.

Codebase Information

- Repository Link: **https://github.com/Harshit-Suyal/FSD-PBL**
- Branch: main

- Important Commits:
  - Initial project setup with folder structure (Frontend + Backend)
  - User authentication system with JWT implementation
  - MongoDB schema integration (users, jobs, applications)
  - REST API development for job posting and applications
  - Frontend dashboard and job listing UI implementation

Testing and Validation Status

Test Type	Status	Notes
Unit Testing (Backend Logic)	Pass	Core functionalities like authentication and job handling tested through code-level checks
Authentication Testing	Pass	Login, signup, and role-based access verified successfully
Database Testing	Pass	Data storage, retrieval, and updates validated using MongoDB
UI Testing	Pass	Interface tested across different screen sizes for responsiveness
Integration Testing	Pass	Some synchronization issues between frontend and backend being refined

End-to-End Testing	Pass	Full user workflow (job posting → applying → tracking) under testing

Deliverables Progress

The core features of the GigConnect platform have been fully completed and are functioning successfully. Frontend development, backend APIs, database integration, and user authentication with role-based access have all been implemented, providing a robust foundation for the system.

The Job Management module has been full developed, including both basic functionalities like job posting and application tracking, as well as advanced features that enhances usability and efficiency.
Key components such as the Rating & Feedback system and Payment Integration have also been successfully implemented, completing all major modules of the platform. Additionally, UI/UX enhancements and performance optimizations have been carried out to improve overall user experience and system performance.

Testing and debugging have been done to ensure smooth integration and reliable functionality across all components.